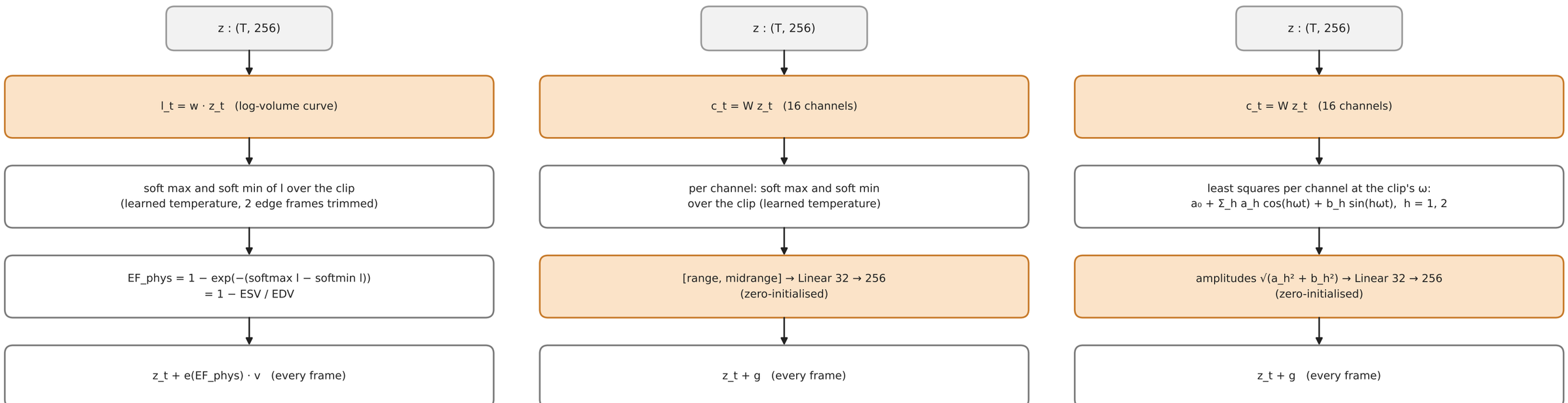
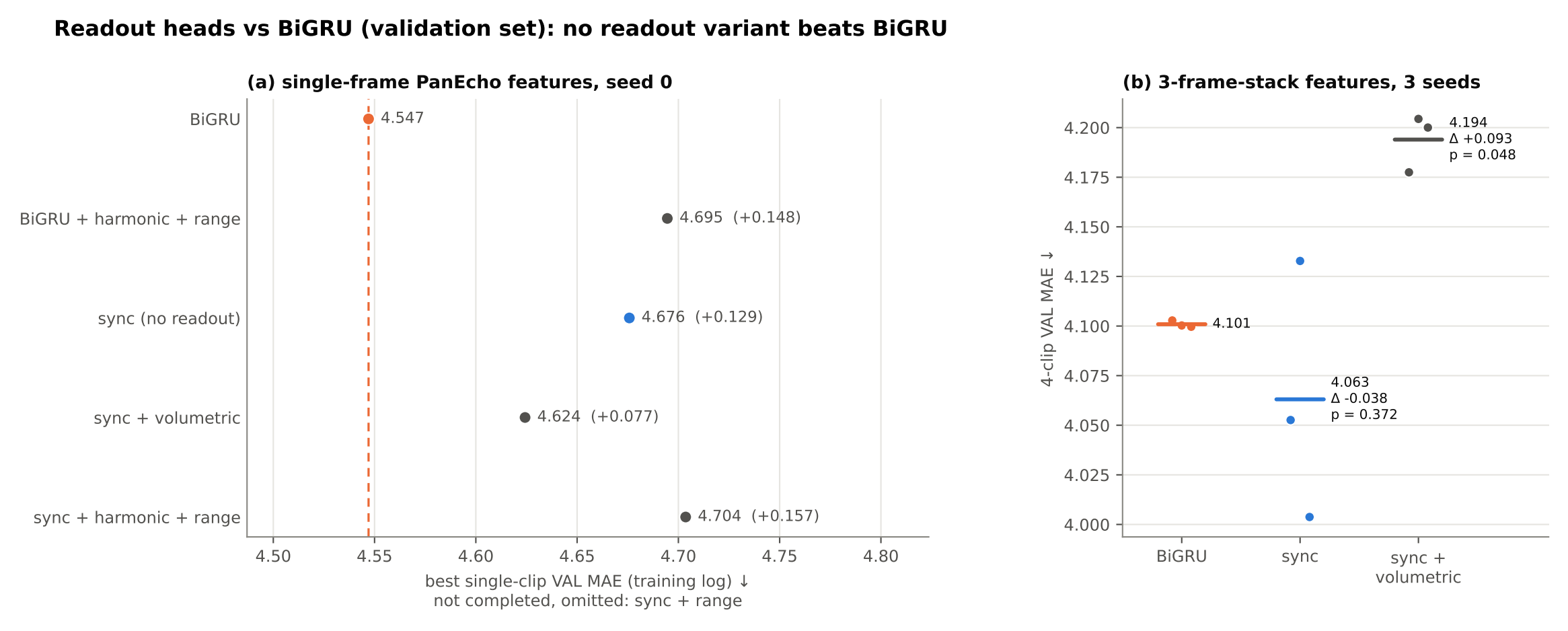
Explored readout heads on the sync block output

trained · no parameters · tensor

Each attaches to the block output z (after the final RMSNorm) and adds a vector to every frame, so the shared mean pool + head can use it. Explored; none was adopted (no VAL gain).




Readout heads vs BiGRU (validation set): no readout variant beats BiGRU
(a) single-frame PanEcho features, seed 0
BiGRU
BiGRU + harmonic + range
sync (no readout)
sync + volumetric
sync + harmonic + range
4.547
4.695 (+0.148)
4.676 (+0.129)
4.624 (+0.077)
4.704 (+0.157)
4.50
4.55
4.60
4.65
4.70
4.75
4.80
best single-clip VAL MAE (training log) ↓
not completed, omitted: sync + range
(b) 3-frame-stack features, 3 seeds
4-clip VAL MAE ↓
4.200
4.175
4.150
4.125
4.100
4.075
4.050
4.025
4.000
4.194
Δ +0.093
p = 0.048
4.101
4.063
Δ -0.038
p = 0.372
BiGRU
sync
sync + volumetric